

1. Let the units for mass, length and time be $\mathcal{M}, \mathcal{L}, \mathcal{T}$ respectively. Then the units for the Planck units are:

- $[\hbar] = [h] = [1 \text{ J} \cdot \text{s}] = [1 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-1}] = \mathcal{M} \mathcal{L}^2 \mathcal{T}^{-1}$
- $[c] = [1 \text{ m} \cdot \text{s}^{-1}] = \mathcal{L} \mathcal{T}^{-1}$
- $[G] = [1 \text{ kg}^{-1} \cdot \text{m}^3 \cdot \text{s}^{-2}] = \mathcal{M}^{-1} \mathcal{L}^3 \mathcal{T}^{-2}$

From this we get

- $[\hbar G] = \mathcal{L}^5 \mathcal{T}^{-3}$
- $[\hbar G / c^3] = \mathcal{L}^2$
- $[\hbar G / c^5] = \mathcal{T}^2$

And therefore we have

- Planck length $l_{pl} = \sqrt{\hbar G / c^3}$
- Planck time $t_{pl} = \sqrt{\hbar G / c^5}$
- Planck mass $M_{pl} = \hbar t_{pl} / l_{pl}^2 = \hbar \sqrt{\hbar G / c^5} / (\hbar G / c^3) = \sqrt{c \hbar / G}$

(ii) Simply substitute in

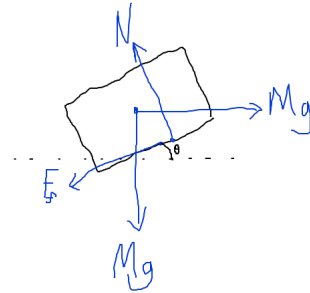
$$M_{pl} = \sqrt{\frac{c \hbar}{G}} = \sqrt{\frac{(2.997 \times 10^8)(6.63 \times 10^{-34} / 2\pi)}{6.67 \times 10^{-11}}} = \mathbf{2.177 \times 10^{-8} \text{ kg}}$$

(iii) We have $E_{grav} = \frac{G m_e^2}{\lambda} = \frac{G m_e^2}{h / m_e c} = \frac{G m_e^3 c}{2\pi \hbar}$ thus

$$\frac{E_{grav}}{m_e c^2} = \frac{G m_e^2}{2\pi \hbar c} = \frac{1}{2\pi} \left(\frac{m_e}{M_{pl}} \right)^2 = \frac{1}{2\pi} \left(\frac{9.11 \times 10^{-31}}{2.177 \times 10^{-8}} \right)^2 = \mathbf{2.787 \times 10^{-46}}$$

(iv) We have $E_{grav} = \frac{G M_{pl}^2}{\lambda} = \frac{G M_{pl}^2}{h / M_{pl} c} = \frac{G M_{pl}^3 c}{2\pi \hbar}$ thus

$$\frac{E_{grav}}{M_{pl} c^2} = \frac{G M_{pl}^2}{2\pi \hbar c} = \mathbf{\frac{1}{2\pi}}$$



2) (i) Let's draw the free-body diagram:

$$F_{net(\parallel)} = F_f + Mg \sin \theta - Mg \cos \theta = 0 \therefore |F_f| = \mathbf{mg |\cos \theta - \sin \theta|}$$

$$F_{net(\perp)} = F_f - Mg \sin \theta - Mg \cos \theta = 0 \therefore |N| = \mathbf{mg (\cos \theta + \sin \theta)}$$

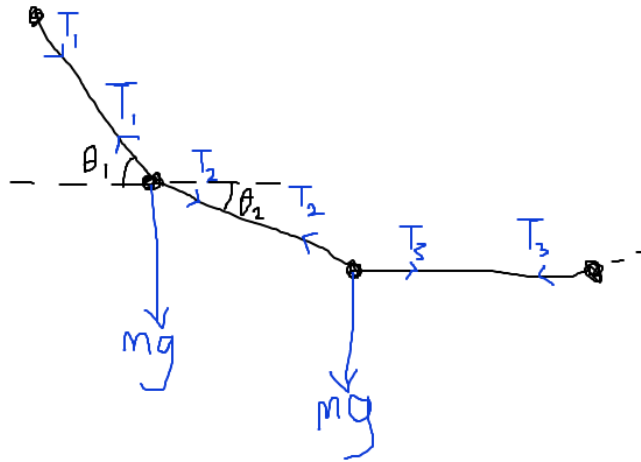
(ii) For block to remain at rest,

$$\frac{|F_f|}{|N|} \leq \mu \therefore \left| \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} \right| \leq \mu \therefore \left| \tan \left(\theta - \frac{\pi}{4} \right) \right| \leq \mu$$

which gives the range of values

$$\mathbf{\frac{\pi}{4} - \arctan \mu \leq \theta \leq \frac{\pi}{4} + \arctan \mu}$$

3) (i) Drawing a diagram (the other half is symmetric):



$F_{net} = 0$ to the first butterfly:

- $T_1 \cos \theta_1 = T_2 \cos \theta_2$
- $T_1 \sin \theta_1 = mg + T_2 \sin \theta_2$

$F_{net} = 0$ to the second butterfly:

- $T_2 \cos \theta_2 = T_3$
- $T_2 \sin \theta_2 = mg$

Therefore we have the following:

- $T_1 \cos \theta_1 = T_2 \cos \theta_2 = T_3$
- $T_1 \sin \theta_1 = 2mg$
- $T_2 \sin \theta_2 = mg$

From this we have

$$T_1 = \frac{2mg}{\sin \theta_1}$$

$$T_3 = T_1 \cos \theta_1 = \frac{2mg \cos \theta_1}{\sin \theta_1} = 2mg \cot \theta_1$$

and

$$T_2 = \sqrt{(T_2 \sin \theta_2)^2 + (T_2 \cos \theta_2)^2} = \sqrt{(mg)^2 + (2mg \cot \theta_1)^2}$$

$$= mg \sqrt{1 + 4 \cot^2 \theta_1}$$

(ii) From the above equations,

$$\theta_2 = \arctan \frac{T_2 \sin \theta_2}{T_2 \cos \theta_2} = \arctan \frac{mg}{2mg \cot \theta_1} = \arctan \left(\frac{1}{2} \tan \theta_1 \right)$$

(iii) We have

$$D = 2l \cos \theta_1 + 2l \cos \theta_2 + l = l \left(2 \cos \theta_1 + 2 \cos \left[\arctan \left(\frac{1}{2} \tan \theta_1 \right) \right] + 1 \right)$$

so $a = b = 2$.

4) We have the kinetic energy of the meteorite when it collides with Earth is

$$K = \frac{1}{2}mv_i^2 + \frac{GM_E m}{R_E}$$

The energy transfer to the meteor in the form of heat is then $Q = \frac{1}{2}K$. We compute this as

$$Q = \frac{1}{2} \left(\frac{1}{2}mv_i^2 + \frac{GM_E m}{R_E} \right) = \frac{1}{2} \left(\frac{1}{2} \cdot 670 \cdot 14000^2 + \frac{(6.67 \times 10^{-11})(6 \times 10^{24})(670)}{6.4 \times 10^6} \right) \\ = 5.378 \times 10^{10} \text{ J}$$

Now the final temperature (assuming it is above the boiling point) satisfies

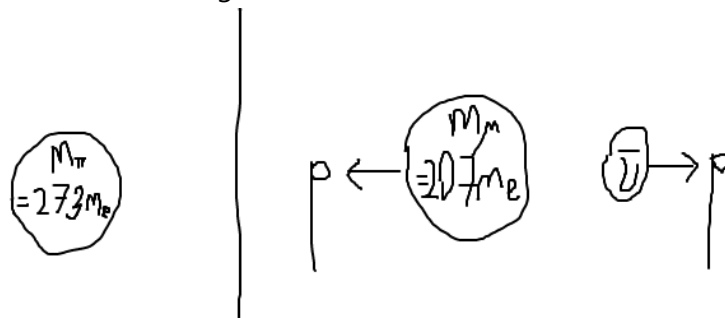
$$Q = mc(T_f - T_i) + mL_{fus} + mL_{vap}$$

thus

$$T_f = T_i + \frac{1}{c} \left(\frac{Q}{m} - L_{fus} - L_{vap} \right) \\ = -15 + \frac{1}{1170} \left(\frac{5.378 \times 10^{10}}{670} - 3.97 \times 10^5 - 1.14 \times 10^7 \right) = \mathbf{58505^\circ\text{C}}$$

which is indeed above the boiling point $2450 \text{ K} = 2177^\circ\text{C}$.

5) Drawing a picture (and noting conservation of momentum):



Now conservation of energy:

$$m_\pi c^2 = \sqrt{(m_\mu c^2)^2 + p^2 c^2} + pc \therefore 273m_e c^2 - pc = \sqrt{207^2 m_e^2 c^4 + p^2 c^2} \\ \therefore 273 - \frac{p}{m_e c} = \sqrt{207^2 + \left(\frac{p}{m_e c}\right)^2} \therefore 273^2 - 2 \cdot 273 \frac{p}{m_e c} + \left(\frac{p}{m_e c}\right)^2 = 207^2 + \left(\frac{p}{m_e c}\right)^2 \therefore p \\ = \frac{5280}{91} m_e c = 58.022 m_e c$$

From this we compute the energies

$$E_{\bar{\nu}} = pc = \frac{5280}{91} m_e c^2 = \frac{5280}{91} (9.11 \times 10^{-31})(2.997 \times 10^8)^2 / 1.6 \times 10^{-19} \\ = \mathbf{2.97 \times 10^7 \text{ eV}}$$

$$K_\mu = E_\pi - E_{\bar{\nu}} - m_\mu c^2 = \left(273 - \frac{5280}{91} - 207 \right) m_e c^2 \\ = \frac{726}{91} (9.11 \times 10^{-31})(2.997 \times 10^8)^2 / 1.6 \times 10^{-19} = \mathbf{4.08 \times 10^6 \text{ eV}}$$

6) We express the (gravitational) potential energy and kinetic energy of the system at small-angle displacements as

$$U = mgR(1 - \cos \theta) \approx mgR \cdot \frac{\theta^2}{2} = \frac{mgR}{2} \theta^2$$

$$K = \frac{1}{2} I \dot{\theta}^2 = \frac{1}{2} \left(\frac{1}{2} MR^2 + \frac{1}{2} mr^2 + mR^2 \right) \dot{\theta}^2$$

We know that for potential and kinetic energies of the form $U = \mu \xi^2$, $K = \kappa \dot{\xi}^2$, the angular frequency of small oscillation is $\omega = \sqrt{\mu/\kappa}$. We will use this for the second part (in determining period). To find speed as it passes through equilibrium position:

$$K = U \therefore \frac{1}{2} \left(\frac{1}{2} MR^2 + \frac{1}{2} mr^2 + mR^2 \right) \left(\frac{v}{R} \right)^2 = mgR(1 - \cos \theta)$$

which simplifies to

$$\frac{1}{4} \left(\frac{M}{m} + \frac{r^2}{R^2} + 2 \right) v^2 = Rg(1 - \cos \theta)$$

$$\therefore v = 2 \sqrt{\frac{Rg(1 - \cos \theta)}{\left(\frac{M}{m} + \left(\frac{r}{R} \right)^2 + 2 \right)}}$$

so $\alpha = \beta = 2$.

Now for the period of motion, one has

$$\omega = \sqrt{\frac{mgR/2}{\frac{1}{2} \left(\frac{1}{2} MR^2 + \frac{1}{2} mr^2 + mR^2 \right)}} \therefore T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{\left(\frac{M}{m} + \frac{r^2}{R^2} + 2 \right) R}{2g}}$$

7) Rotational mechanics:

$$\tau = T_{top}r - T_{bottom}r = (T_{top} - T_{bottom})r = I\alpha = \frac{1}{2}MR^2\alpha$$

$$\therefore T_{bottom} = T_{top} - \frac{1}{2}MR^2\alpha/r = 135 - \frac{1}{2}(80.0) \left(\frac{1.25}{2} \right)^2 (1.67)/(0.230)$$

$$= \mathbf{21.55 \text{ N}}$$

8) (i) The wave acquires a π -rad phase shift when reflecting off the flat glass plate ($n_{\text{glass}} > n_{\text{air}}$), but not off the concave surface ($n_{\text{air}} < n_{\text{glass}}$). The path difference in the center is thus the optical path length

$$t = 50\lambda = 50 \cdot 589 = 29450 \text{ nm} = \mathbf{29.45 \mu\text{m}}$$

(ii) The radius of the outermost dark ring is computed by Pythagoras,

$$r = \sqrt{R^2 - (R - t)^2} = \sqrt{8.00^2 - (8.00 - 29.45 \times 10^{-6})^2} = 0.0217 \text{ m} \\ = \mathbf{21.7 \text{ mm}}$$

(iii) The focal length of the lens is given by the lensmaker equation

$$\frac{1}{f} = (n - 1) \left(-\frac{1}{R} \right) \therefore f = -\frac{R}{n - 1} = -\frac{8.00}{1.50 - 1} = \mathbf{-16.0 \text{ m}}$$

Where the negative sign of the focal length means that the lens is divergent.

9) (a) The magnetic field inside the toroid, by the approximation, is uniformly

$$B = \mu_0 n I = \mu_0 \left(\frac{N}{2\pi R} \right) I$$

and the total flux through the toroid is

$$\phi = NBA = \frac{1}{2\pi} \mu_0 \frac{N^2 A}{R} I$$

therefore the induced emf is

$$\varepsilon = -\frac{d\phi}{dt} = -\frac{1}{2\pi} \mu_0 \frac{N^2 A}{R} \cdot \frac{dI}{dt} = -L \frac{dI}{dt} \therefore L = \frac{1}{2\pi} \mu_0 \frac{N^2 A}{R}$$

which means $\kappa = 1/2\pi, \alpha = 2$ are the desired constants.

(b) (i) We apply Ampere's Law to a closed circular loop of radius r within the solenoid:

$$\oint \vec{B} \cdot d\vec{l} = B_r(2\pi r) = \mu_0 I_{\text{total}} = \mu_0 N I \therefore B_r = \frac{\mu_0 N I}{2\pi r}$$

Thus the total flux through the toroid is

$$\phi = N \iint B dA = N \int_0^h \int_a^b B_r dr dh = N \left(\int_0^h dh \right) \left(\int_a^b \frac{\mu_0 N I}{2\pi r} dr \right) = \frac{\mu_0 N^2 h I}{2\pi} \ln \frac{b}{a}$$

therefore the induced emf is

$$\varepsilon = -\frac{d\phi}{dt} = -\frac{1}{2\pi} \mu_0 N^2 h \ln \frac{b}{a} \cdot \frac{dI}{dt} = -L' \frac{dI}{dt} \therefore L' = \frac{1}{2\pi} \mu_0 N^2 h \ln \frac{b}{a}$$

which means $\kappa' = R/2\pi, \beta = 2$ are the desired constants.

(ii) We suppose the approximation was achieved by taking $R = (a + b)/2$, $A = h(b - a)$. The percentage error of the approximation is

$$\begin{aligned}\frac{L - L'}{L'} &= \frac{\frac{h(b-a)}{(a+b)/2} - h \ln \frac{b}{a}}{h \ln \frac{b}{a}} = \frac{2(b-a)}{(a+b) \ln \frac{b}{a}} - 1 = \frac{2(12.0 - 10.0)}{(12.0 + 10.0) \ln \frac{12.0}{10.0}} - 1 \\ &= 0.997 - 1 = -0.003 = -\mathbf{0.3\%}\end{aligned}$$

10) At rest, the total energy of the box (its rest energy) and the photons (each has energy $h\nu$) is

$$E = Mc^2 + N h \nu = M_{\text{sys}} c^2$$

where $M_{\text{sys}} c^2$ is the rest mass of the system. Thus

$$M_{\text{sys}} = \mathbf{M} + N \frac{h\nu}{c^2}$$